

NAKUL KUMBRIA

Oceanside, CA | Software Development Engineer / Backend Engineer | (714) 924-0481 | nakul.kumbria.uni@outlook.com | LinkedIn | GitHub

EDUCATION

University of Southern California
Master of Science in Computer Science

January 2025-Present

University of Texas Permian Basin
Bachelor of Science in Computer Science
Dean's List | President's Honor Roll (3.851 GPA)

January 2018-December 2022

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, PHP, JavaScript
Frontend, Backend & Frameworks: Spring Boot, Node.js, Laravel, GraphQL, React, HTML, CSS, Bootstrap
Databases: MySQL, SQL Server, MongoDB, Redshift
Cloud & Data: AWS (EC2, S3, Cloud9, SageMaker), Databricks, PySpark
Tools: Git, Selenium, Power BI, Tableau, Jira, Salesforce
Concepts: Data Structures & Algorithms, REST APIs, Cloud Architecture, Automation, System Design

PROFESSIONAL EXPERIENCE

Software Engineer / IT Project Manager, Enterprise Technology Operations

November 2024-Present

Thermo Fisher Scientific, Carlsbad, CA

- Engineered an AI-powered chatbot using Python, OpenAI API, LangChain, and Streamlit to answer natural-language questions across internal datasets and reduce dependency on manual knowledge lookup.
- Implemented Retrieval-Augmented Generation (RAG) pipelines in Databricks, improving response accuracy and reducing manual knowledge lookup time.
- Built and maintained cloud-native applications using AWS EC2, S3, and Cloud9 with scalable, modular architectures
- Devised Python-based automation services to extract, process, and visualize Jira Agile metrics, generating sprint analytics and automated stakeholder email reports that improved engineering visibility into delivery trends.
- Collaborated on cloud cost optimization by analyzing EC2 usage patterns, identifying underutilized resources, and contributing to measurable infrastructure cost savings.
- Administer Databricks non-production environments, managing clusters, access control, data pipelines, and workspace reliability.

Systems Analyst, Enterprise Technology Operations

June 2022-November 2024

(Software & Data Engineering Focus) at Thermo Fisher Scientific, Carlsbad, CA

- Built and optimized ETL pipelines using AWS S3, Databricks, Redshift, and SQL, enabling large-scale financial and sales analytics.
- Engineered Python automation scripts to ingest data from SharePoint and Salesforce into SQL Server, reducing manual data handling.
- Created data visualization dashboards using Power BI, Tableau, and Salesforce, supporting executive-level financial decision-making.
- Automated operational workflows using UiPath, Power Automate, and Selenium, delivering significant time and cost savings.
- Managed and queried relational databases using SQL Server, DBeaver, MySQL, ensuring data accuracy and performance.

PROJECTS

Art Collection Explorer | Python, Node.js, Express, Google Cloud Run, REST APIs, Los Angeles

April 2026-May 2026

- Developed and deployed a cloud-hosted full-stack artwork discovery platform integrating Harvard Art Museums and Metropolitan Museum of Art APIs using Node.js, Express, Docker, and Google Cloud Run.
- Built backend API aggregation services supporting filtering, pagination, caching, and asynchronous external API requests for scalable real-time data retrieval.
- Designed responsive frontend features that includes dynamic search, favorites management, and real-time result rendering optimized for performance and user experience.

Movie Explorer, Movie Explorer with Reviews | Python, Flask, Google Cloud Run, Docker, Los Angeles

March 2026-April 2026

- Engineered and deployed a full-stack movie discovery platform using Flask, REST APIs, Docker, NGINX, and Google Cloud Run to aggregate movie metadata, trailers, ratings, and reviews
- Designed modular backend services for API orchestration, scalable request handling, secure configuration management, and real-time content retrieval across external data sources
- Built responsive frontend workflows with dynamic routing, movie search, and detailed content views optimized for usability, performance, and cloud-based application delivery

INTERNSHIP EXPERIENCE

Web Developer

June 2021-August 2021

Experis IT (Manpower Group), Gurugram

- Developed backend components using PHP (CodeIgniter) and MySQL for enterprise web applications.
- Designed relational database schemas focused on performance, reliability, and security.
- Collaborated with cross-functional teams to troubleshoot performance issues and deliver scalable solutions.

Web Developer

December 2020-May 2021

SUNLOC, Delhi

- Built and maintained full-stack web applications using PHP, MySQL, HTML, CSS, JavaScript.
- Implemented database security, backup, and recovery mechanisms.